

Discussion for ***When Learning Increases
Uncertainty: Consumer Evaluation
under Multidimensional Uncertainty***

by W. Jason Choi (Maryland) and Jiwoong Shin (Yale)

Discussant: Jesse Yao (CUHK)

An Example



Q: Predict if Jason will be tenured this fall.

An Example



Q: Predict if Jason will be tenured this fall.

Research: more certain belief

An Example



Q: Predict if Jason will be tenured this fall.

Research: more certain belief

Teaching: even more certain

An Example



Q: Predict if Jason will be tenured this fall.

Research: more certain belief

Teaching: even more certain

Service: more and more certain

An Example



Q: Predict if Jason will be tenured this fall.

Research: more certain belief

Teaching: even more certain

Service: more and more certain

Learning makes a DM more certain in their belief

An Example



Q: Predict if Jason will be tenured this fall.

BUT...

An Example



Things are different if there is uncertainty about university finance

- Worst case: no one can be promoted
- Best case: can afford to promote anyone sufficiently good

An Example



- Does not matter if the candidate is weak (low chance in either case)
- Matters a lot if the candidate is good (low chance in one case/high chance in the other case)
 - could lead to a higher uncertainty after learning that Jason is so good

Fast forward...



**Congrats on getting
tenured at Maryland!**

Fantastic researcher & department

Key Mechanism

- (page 16) Resolving one dimension of uncertainty can make overall value more sensitive to the unresolved uncertainty.

Key Mechanism

$$\theta = \omega\alpha + (1 - \omega)\beta$$

1. If $1 - \omega$ is larger, then the unknown value β plays a larger role in determining the state θ

Before learning: $\alpha = 10$, $\beta = 0$ or 10 with equal probability

- If the DM learns that $\omega = 1$, then $\theta = 10 \rightarrow \beta$ does not matter (no remaining uncertainty)
- If the DM learns that $\omega = 0$, then θ can be 0 or $10 \rightarrow \beta$ matters a lot (high remaining uncertainty)

Key Mechanism

$$\theta = \omega\alpha + (1 - \omega)\beta$$

2. If α and β are more different, then the unknown weight ω plays a larger role in determining the state θ

Before learning: $\alpha = 10$, $\beta = 0$ or 10 with equal probability

- If the DM learns that $\beta = 10$, then $\theta = 10 \rightarrow \omega$ does not matter (no remaining uncertainty)
- If the DM learns that $\beta = 0$, then θ can be 0 or 10 $\rightarrow \omega$ matters a lot (high remaining uncertainty)

Some Clarifications

$$\theta = \omega\alpha + (1 - \omega)\beta$$

1. The asymmetry between two attribute (α is known, β is unknown) is not a restriction

The current setup is the simplest one that makes the key mechanism work (one unknown attribute value, and unknown weight)

Some Clarifications

$$\theta = \omega\alpha + (1 - \omega)\beta$$

2. The additive functional form is not a restriction

- What is really needed is some **interaction/correlation** between different components of the state
- Multiple dimensions per se are not enough: $\theta = \sum_i \omega_i \beta_i$

Some Clarifications

$$\text{Adoption payoff: } \theta - p - \frac{k}{2}(x - \theta)^2$$

3. The quadratic functional form of the calibration loss is not a restriction

The current setup leads to a simple closed-form for the optimal action $x^* = E[\theta|I]$

Some Clarifications

4. Learning increases uncertainty means the ex-post conditional variance (not the ex-ante expected conditional variance) may exceed the ex-ante unconditional variance

The DM may be more uncertain after some signal realizations, but the expected uncertainty after learning always decreases

Some Clarifications

5. The DM may be more uncertain after some signal realizations occurs in other learning models.

Some Clarifications

Discrete-time model:

$$\theta \in \{0,1\}, P(s = h|\theta = 1) = P(s = l|\theta = 0) = q > \frac{1}{2}$$

- When the prior $P(\theta = 1)$ is high and q is close to $\frac{1}{2}$, a $s = l$ signal can move the belief closer to $\frac{1}{2} \Rightarrow$ more uncertainty

Some Clarifications

Continuous-time model:

$$\theta \in \{0,1\}, dS_t = \theta dt + \sigma dW_t \text{ (} W_t \text{: white noise)}$$

- When the prior $P(\theta = 1)$ is high and there is a large negative white noise, the signal can move the belief closer to $\frac{1}{2} \Rightarrow$ more uncertainty

Some Clarifications

Key difference: the existing literature that generates such results features signals about the overall valuation

(the posterior mean moves towards the middle)

This paper: signals about specific attributes/weights

Learning one dimension changes the payoff relevance of other unresolved dimensions

(the posterior mean may be closer to the extreme)

- Direct **economic mechanism**
- Much richer space for **intervention**

Firm Intervention

Current paper: firm disclosure as a way of changing the search order/encouraging search

Firm prefers consumers to have higher adoption value

→ firm wants consumers to have lower residual uncertainty

→ discloses what consumers would **not** have learned

Very clean intuition!

Firm Intervention

Current paper: firm disclosure as a way of changing the search order/encouraging search

Q: (search deterrence) Whether a firm may have incentive to deter search?

→ Make some attributes harder to learn

(may need to jointly consider the pricing decision)

Regulator Intervention

Current paper: firm disclosure as a way of changing the search order/encouraging search

Q: (regulator intervention) What if a regulator rather than the firm is designing the intervention?

Empirical Implications

The uncertainty amplification can lead to interesting testable predictions

Challenge: multiple objects are moving at the same time given new information (e.g., mean and variance)

Moving Beyond This Paper

The DM may be more uncertain after some signal realizations, but the expected uncertainty after learning always decreases (otherwise the DM will not learn)

Q: Is it possible that the expected uncertainty after learning may increase under some circumstances?

Mathematically impossible without considering strategic effects;

Competition, endogenous pricing, etc. may play an important role...